

1) Prompt Design (from the Clean CSV)

Goals:

- Convert your structured CSV data into a format that can be inserted directly into prompts.
- Create question-answer (QA) examples and instructions to guide the LLM's output.

Key Tasks:

1. Extract Relevant Fields

- Identify the columns you'll showcase in your prompts (e.g., Funder Name, Focus/Theme, Funding Range, Eligibility).
- Summarize or compress large text fields (like "Additional Notes") into concise bullet points so they fit within prompt length limits.

2. Build "Context Blocks"

- For each funder entry, create a short snippet that includes the essential CSV details in a user-friendly format.

Example snippet:

Funder: Walt Disney Conservation Fund
Type: Corporate Grant
Focus: Wildlife Conservation, Community Engagement
Funding Range: Up to \$50,000
Application Timeline: Annual (spring)
Eligibility: Nonprofits aligned with wildlife protection

These become building blocks you'll embed into your prompts as needed.

3. Craft Few-Shot Examples

- Write out a handful of typical questions a user might ask, and show the "ideal" answer that references the CSV data.

- This helps the LLM see exactly how to respond.

4. Create “System” or “Instruction” Prompts

- E.g., “You are an expert on biodiversity grants. You have access to the following information. Use only the data provided to answer the user’s questions. If something isn’t in the data, say ‘I don’t have that information.’”

2) Prompt Testing

Goals:

- Ensure the prompts reliably return correct, structured information from the CSV.
- Identify and fix any issues (hallucinations, missing references, etc.).

Key Tasks:

1. Small-Scale Tests

- Manually insert one or two “Context Blocks” (from your CSV) into the prompt and pose sample queries (e.g., “Which funder supports marine biodiversity?”).
- Check if the LLM properly references the data.

2. Iterate and Refine

- Adjust the wording of your instructions or examples if the LLM yields incomplete or inaccurate answers.
- For instance, if the LLM misses the “Funding Range,” explicitly highlight it in your context or add a line in your instructions (“Always cite the Funding Range if available”).

3. Document Best Practices

- Keep a short doc or table of “Prompt Variation” → “LLM Output” so you can see which prompt structures yield the best accuracy.

3) Prompt Integration

Goals:

- Integrate the refined prompts into your chatbot or app pipeline.
- Automate how your system selects and inserts relevant CSV data into the LLM prompt.

Key Tasks:**1. Automate Context Retrieval (Optional Basic Approach)**

- For now, you might do a simple keyword match on the user's query and pull the relevant row(s) from your CSV.
- Insert these "Context Blocks" into the final user prompt before sending it to the LLM.

2. Design the "Conversation Flow"

- Decide how you'll handle user follow-up questions. If a user asks for more details, your system might re-insert the same context block with an updated user query.
- This ensures the LLM always has the CSV data at hand without repeatedly uploading the entire CSV.

3. Set Up Logging

- Log user queries and the LLM's responses for further analysis and iteration (especially if you plan to eventually fine-tune or refine prompts).

4) LLM Test

Goals:

- Evaluate the final integrated solution across multiple queries or even multiple LLMs.
- Confirm that the chatbot accurately references the CSV data and meets your performance needs.

Key Tasks:

1. Create a Test Suite

- A small set of standardized queries (e.g., 5–10 real-world questions) that you'll run to compare LLM outputs.
- Example queries:
 - "Which funder provides multi-year grants for ocean conservation?"
 - "Do any funders accept rolling applications for tropical rainforest projects?"
 - "Show me corporate grants that focus on wildlife and education."

2. Compare LLMs (Optional)

- If you're testing multiple models (e.g., GPT-4, Claude, or a Llama variant), run the same prompts and note differences in accuracy, detail, or cost.

3. Collect Feedback

- If possible, have domain experts or end users evaluate the LLM's answers for correctness, clarity, and helpfulness.
- Document any areas where the model still hallucinated or omitted data.

Prompt Design Examples

Here are **sample prompts** and how you might embed your CSV data for a biodiversity funding query:

Example 1: Single Funder Reference

System/Instruction Prompt:

"You are a knowledgeable assistant with details about biodiversity grants from a CSV file."

You can only use the provided data to answer questions. If the answer isn't in the data, say:

'I'm sorry, but I don't have that information.'

When you mention a funder, please refer to them by name as it appears in the data."

Context Block:

Funder: Walt Disney Conservation Fund

Type: Corporate Grant

Focus: Wildlife Conservation, Community Engagement

Funding Range: Up to \$50,000

Application Timeline: Annual (spring)

Eligibility: Nonprofits aligned with wildlife protection

User Prompt:

"Which grant focuses on wildlife and education?"

Ideal LLM Answer:

The Walt Disney Conservation Fund focuses on wildlife conservation and often supports education-related aspects. They provide grants up to \$50,000, and accept applications annually in the spring.

Example 2: Multiple Funder Comparison

System/Instruction Prompt:

"You have data about various funders:

- 1) David and Lucile Packard Foundation (Private Foundation, focus on biodiversity, ocean conservation, \$50,000+ rolling)
- 2) Oak Foundation (Private Foundation, focus on biodiversity, climate change, \$25,000-\$500,000, open)
- 3) Shell Foundation (Corporate, focus on energy & social impact with biodiversity co-benefits, variable timeline)

The user may ask you about comparing these. Give short, factual answers from the data."

User Prompt:

"Which of these foundations offer larger grants for biodiversity?"

Ideal LLM Answer:

Based on the provided data, the Oak Foundation can offer grants ranging from \$25,000 to \$500,000, which could exceed the typical \$50,000+ range of the David and Lucile Packard Foundation.

Shell Foundation's funding range varies widely, but it primarily focuses on energy and social impact, with biodiversity as a secondary theme.

Example 3: Field-Specific Answer

System/Instruction Prompt:

"You are an expert on biodiversity grants. You have access to details like Funder Name, Funder Type, Funding Range, and Geographic Focus. Answer precisely using these data points."

Context Block:

Funder: Rainforest Trust

Type: Private Foundation

Focus: Rainforest Conservation, Protected Areas

Funding Range: \$5,000+

Geographic Focus: Tropical Regions

Timeline: Rolling

Eligibility: Nonprofits establishing or expanding protected areas

User Prompt:

"Is Rainforest Trust a good option for protecting tropical habitats year-round?"

Ideal LLM Answer:

Yes. Rainforest Trust focuses on rainforest conservation and protected areas, with a particular emphasis on tropical regions. Their funding timeline is rolling, and they support nonprofits working to establish or expand protected areas.